

Security+ 3.1 · Other Infrastructure Concepts

Master CyberSec — Official CompTIA Security+ (SY0-701) Comprehensive Study Guide · **Lesson 3.1 — Other Infrastructure Concepts (14:24)** · [Official YouTube Video ↗](#) · [Professor Messer Study Hub ↗](#)

Executive Lesson Summary: Infrastructure types: VMs use a hypervisor to virtualize hardware, providing strong isolation; containers (e.g., Docker) share the host kernel, are lightweight, but have weaker default isolation. ICS/OT (Industrial Control Systems/Operational Technology) include PLCs controlling factories and SCADA for monitoring; they are extremely patch-resistant, often air-gapped, with USB as a common attack vector, and use protocols like Modbus and DNP3. IoT/embedded devices run RTOS and represent a large, often forgotten attack surface.

High-Yield Core Concepts Tested on Exam

- VM: strong isolation via hypervisor; container: lightweight, shares host kernel
- ICS/OT (factories, utilities): very difficult to patch — air gap, industrial DMZ
- IoT/embedded devices: huge attack surface, weak patching
- PLC controls the process; SCADA monitors and supervises

Exam Tips & Classic Scenarios

- 'isolated with strong boundary' = VM; 'shares kernel with host' = container
- 'industrial system isolated with special protocols' = ICS/OT
- 'USB is a common attack vector in air-gapped environments' = ICS/OT

Typical Exam Scenario: An industrial plant has a PLC connected to the business network. This represents an ICS/OT risk and requires strict isolation.

Beyond the Video: Official CompTIA Exam Concepts, Protocols & Traps (Lesson 3.1 — Other Infrastructure Concepts)

Why is this advanced coverage required for SY0-701? Video overviews provide foundational familiarity, but CompTIA tests practical engineering judgment and requires deep familiarity with the following protocols, acronyms, and operational nuances in this domain:

- **Principle of Least Privilege:** Users and computational entities must be granted only the minimum permissions strictly required to perform designated tasks.
- **Defense-in-Depth (Layered Security):** Application of multiple security controls (technical, administrative, physical) to prevent single-point perimeter compromise.

Classic CompTIA Traps & Question Framing:

- Avoid rote memorization: CompTIA exam prompts regularly adjust technical constraints requiring you to defend why one valid security control is the *MOST appropriate* or *FIRST step*.
- Scrutinize question qualifiers: 'without additional licensing costs' favors native OS tools; 'granular administrative command accounting' favors TACACS+ over RADIUS; 'non-repudiation' mandates asymmetric private key signing.

Practice Questions with In-Depth Solutions (10 Questions)

Detailed answer key with step-by-step technical rationale and comprehensive distractor breakdown provided below. Attempt answering independently first.

1. Physically isolate a critical system from the network (with no connection at all):

- A)** Air gap
- B)** VLAN
- C)** VPN
- D)** NAT

2. Corporate scenario (Other Infrastructure Concepts): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.1), an analyst evaluates the following technical requirement: 'VM: STRONG isolation (hypervisor). Container: lighter, shares kernel'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Implement the corresponding solution: VM: STRONG isolation (hypervisor). Container: lighter, shares kernel.
- D)** Allow shared access among administrators using generic credentials

3. Corporate scenario (Other Infrastructure Concepts): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.1), an analyst evaluates the following technical requirement: 'ICS/OT (factory, power plant): patching VERY difficult — air gap, industrial DMZ'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to speed up system throughput
- B)** Implement the corresponding solution: ICS/OT (factory, power plant): patching VERY difficult — air gap, industrial DMZ...
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

4. Corporate scenario (Other Infrastructure Concepts): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.1), an analyst evaluates the following technical requirement: 'IoT/devices: HUGE attack surface and WEAK patching'. What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: IoT/devices: HUGE attack surface and WEAK patching...
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators with generic credentials

5. Corporate scenario (Other Infrastructure Concepts): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.1), an analyst evaluates the following technical requirement: 'isolated with strong boundary' = VM; 'shares kernel with host' = container. What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: 'isolated with strong boundary' = VM; 'shares kernel with host' = container
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators with generic credentials

6. Corporate scenario (Other Infrastructure Concepts): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.1), an analyst evaluates the following technical requirement: 'isolated industrial system with special protocol' = ICS/OT'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators using generic credentials
- D)** Implement the corresponding solution: 'isolated industrial system with special protocol' = ICS/OT...

7. Corporate scenario (Other Infrastructure Concepts): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.1), an analyst evaluates the following technical requirement: 'Scenario: industrial plant with PLC connected to the business network = ICS/OT risk — mandatory isolation.' What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: Scenario: industrial plant with PLC connected to the business network = ICS/OT ...
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators with generic credentials

8. Corporate scenario (Other Infrastructure Concepts): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.1), an analyst evaluates the following technical requirement: 'VM: STRONG isolation (hypervisor). Container: lighter, shares kernel.' What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: VM: STRONG isolation (hypervisor). Container: lighter, shares kernel.
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

9. Corporate scenario (Other Infrastructure Concepts): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.1), an analyst evaluates the following technical requirement: 'ICS/OT (factory, power plant): patching VERY difficult — air gap, industrial DMZ.' What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators with generic credentials
- D)** Implement the corresponding solution: ICS/OT (factory, power plant): patching VERY difficult — air gap, industrial DMZ...

10. Corporate scenario (Other Infrastructure Concepts): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.1), an analyst evaluates the following technical requirement: 'IoT/devices: HUGE attack surface and weak PATCHING'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Implement the corresponding solution: IoT/devices: HUGE attack surface and weak PATCHING.
- D)** Allow shared access among administrators using generic credentials

COMPREHENSIVE ANSWER KEY & DISTRACTOR BREAKDOWN

1) A — Air gap.

✓ **Why it is correct:** Air gap (or physical isolation) is the practice of physically separating a critical system from any network, including the Internet, LAN, and WAN, so that there is no physical or logical connection. This prevents remote attacks and data exfiltration over the network. It is used in high-security environments such as industrial control systems (ICS/SCADA), military networks, and voting systems. According to SY0-701 objective 3.3 (security architecture) and 4.1 (security controls), air gap is the strongest isolation measure. The question describes exactly isolating a critical system from the network with no connection, which is the definition of air gap.

✗ **Why the others are wrong:**

- **VLAN:** VLAN (Virtual Local Area Network) is a logical segmentation of a physical network, created on switches, that allows separating broadcast domains and applying security policies. Although it improves logical isolation, it is not physical isolation; devices in different VLANs still share the same physical infrastructure and can be interconnected by routing or misconfiguration. The question requires PHYSICAL isolation with no connection, which VLAN does not provide, as it maintains underlying logical and physical connectivity.
- **VPN:** VPN (Virtual Private Network) is an encrypted tunnel that allows secure communication over untrusted networks, such as the Internet. It provides confidentiality and integrity, but does not physically isolate the system; on the contrary, it connects the system to a remote network securely. The question asks for physical isolation with no connection, which is incompatible with VPN, which presupposes network connectivity.
- **NAT:** NAT (Network Address Translation) is a technique that translates private IP addresses to public ones (and vice versa), allowing multiple devices to share a single public IP address. It hides the internal topology but does not physically isolate; the system remains connected to the network. The question requires physical isolation with no connection, which NAT does not offer, as it is a routing mechanism that maintains connectivity.

2) C — Implement the corresponding solution: VM: STRONG isolation (hypervisor).

Container: lighter, shares kernel..

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.1 (Other Infrastructure Concepts): VM: STRONG isolation (hypervisor). Container: lighter, shares kernel. VMs provide strong isolation via the hypervisor, while containers share the host OS kernel, offering less isolation but greater efficiency. Security implications include container escape risks and the need for kernel hardening.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to incident detection and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user behavior and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

3) B — Implement the corresponding solution: ICS/OT (factory, power plant): patching VERY difficult — air gap, industrial DMZ....

✅ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.1 (Other Infrastructure Concepts): ICS/OT (factory, power plant): patching VERY difficult — air gap, industrial DMZ. Industrial Control Systems (ICS) and Operational Technology (OT) often run legacy systems that cannot be easily patched due to operational constraints. The appropriate control is to implement the corresponding solution, which includes network segmentation (air gap, industrial DMZ), strict access controls, and continuous monitoring to compensate for the inability to patch.

❌ **Why the others are wrong:**

- *Temporarily disable audit logs to speed up system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to incident detection and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user behavior and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

4) A — Implement the corresponding solution: IoT/devices: HUGE attack surface and WEAK patching....

✅ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.1 (Other Infrastructure Concepts): IoT/devices: HUGE attack surface and WEAK patching. IoT devices often have limited security controls and infrequent updates, making them a significant risk.

❌ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detect incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user behavior, and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

5) A — Implement the corresponding solution: 'isolated with strong boundary' = VM; 'shares kernel with host' = container.

✅ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.1 (Other Infrastructure Concepts): 'isolated with strong boundary' = VM; 'shares kernel with host' = container. VMs provide strong isolation via hypervisor, while containers share the host OS kernel, offering less isolation but greater efficiency.

❌ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detect incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user behavior, and do not block automated attacks, injections, or malware at runtime.

- *Allow shared access among administrators with generic credentials*: Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

6) D — Implement the corresponding solution: 'isolated industrial system with special protocol' = ICS/OT....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.1 (Other Infrastructure Concepts): 'isolated industrial system with special protocol' = ICS/OT. Industrial Control Systems (ICS) and Operational Technology (OT) often use proprietary protocols and are isolated from traditional IT networks to protect critical infrastructure. Security measures must account for their unique constraints and availability requirements.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput*: This directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls that rely solely on user behavior and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators using generic credentials*: This destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

7) A — Implement the corresponding solution: Scenario: industrial plant with PLC connected to the business network = ICS/OT

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.1 (Other Infrastructure Concepts): Scenario: industrial plant with PLC connected to the business network = ICS/OT risk — mandatory isolation.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials*: Destroys individual accountability and non-repudiation. Makes it impossible to prove who made critical changes or malicious commands.

8) A — Implement the corresponding solution: VM: STRONG isolation (hypervisor).

Container: lighter, shares kernel..

The correct answer is to implement the corresponding solution: understand that VMs provide strong isolation via the hypervisor, while containers are lighter but share the host kernel. According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.1 (Other Infrastructure Concepts), this distinction is critical for security architecture: hypervisor-based VMs offer stronger isolation boundaries, whereas container escapes can compromise the shared kernel. Distractor breakdown: • *Temporarily disable audit logs to accelerate system throughput*: Incorrect. This directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to incident detection and breaks compliance standards (PCI DSS, ISO 27001). • *Replace preventive technical controls with informal terms of adherence*: Incorrect. Informal terms are administrative

controls that rely solely on user behavior and do not block automated attacks, injections, or runtime malware. • Allow shared access among administrators using generic credentials: Incorrect. This destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

9) D — Implement the corresponding solution: ICS/OT (factory, power plant): patching VERY difficult — air gap, industrial DMZ....

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.1 (Other Infrastructure Concepts): ICS/OT (factory, power plant): patching VERY difficult — air gap, industrial DMZ.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

10) C — Implement the corresponding solution: IoT/devices: HUGE attack surface and weak PATCHING..

✓ **Why it is correct:** Per the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.1 (Other Infrastructure Concepts): IoT/devices: HUGE attack surface and weak PATCHING. IoT (Internet of Things) devices — smart sensors, cameras, thermostats, industrial controllers — dramatically expand the attack surface because they are numerous, often internet-exposed, and frequently deployed with default credentials. Many lack a supported patching mechanism, receive infrequent firmware updates, or have no update path at all, making them persistent vulnerabilities. Security+ emphasizes segmenting IoT devices, changing default credentials, and applying compensating controls when patching is not feasible.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to incident detection and breaches compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user conduct and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

